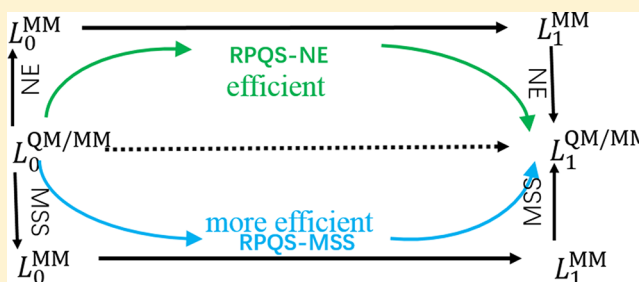


Predicting Relative Binding Affinity Using Nonequilibrium QM/MM Simulations

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Supporting Information

ABSTRACT: Calculating binding free energies with quantum-mechanical (QM) methods is notoriously time-consuming. In this work, we studied whether such calculations can be accelerated by using nonequilibrium (NE) molecular dynamics simulations employing Jarzynski's equality. We studied the binding of nine cyclic carboxylate ligands to the octa-acid deep-cavity host from the SAMPL4 challenge with the reference potential approach. The binding free energies were first calculated at the molecular mechanics (MM) level with free energy perturbation using the generalized Amber force field with restrained electrostatic potential charges for the host and the ligands. Then the free energy corrections for going from the MM Hamiltonian to a hybrid QM/MM Hamiltonian were estimated by averaging over many short NE molecular dynamics simulations. In the QM/MM calculations, the ligand was described at the semiempirical PM6-DH+ level. We show that this approach yields MM → QM/MM free energy corrections that agree with those from other approaches within statistical uncertainties. The desired precision can be obtained by running a proper number of independent NE simulations. For the systems studied in this work, a total simulation length of 20 ps was appropriate for most of the ligands, and 36–324 simulations were necessary in order to reach a precision of 0.3 kJ/mol.



INTRODUCTION

Accurate predictions of the binding affinities of small molecules to biological macromolecules is critical in computational structure-based drug design and is one of the essential challenges of computational chemistry.^{1–3} In order to make these calculations more feasible, numerous free energy calculation methods have been suggested over the past few decades.^{4,5} According to large-scale benchmark tests, alchemical free energy perturbation (FEP) calculations yield rather accurate predictions of relative binding free energies for many protein–ligand complexes, with a mean absolute deviation of 4–6 kJ/mol, but for some proteins the results are poor.^{6–8} The accuracy of such calculations is in principle limited only by the amount of sampling (length of the simulation) and the accuracy of the energy function, which normally is a nonpolarizable force field.^{9–12}

However, nonpolarizable force fields lack an explicit description of electronic response to the specific chemical environment, and the predictive capability of force fields is limited.^{13,14} On the other hand, quantum-mechanical (QM) methods can provide an accurate description of the potential energy surface. Unfortunately, full QM calculations of protein–ligand binding affinity are not feasible because of the steep computational scaling of CPU time and memory

requirements. Even with the hybrid QM/MM scheme,^{15–20} in which the environment is treated at the molecular mechanics (MM) level, it is still very expensive to generate ensembles of structures for a free energy calculation.

The free energy difference between two states is independent of the path of transformation. Therefore, there has recently been much interest in the reference potential scheme, which was pioneered by Gao, Warshel, and co-workers,^{21–25} and then implemented and enhanced by many other groups.^{26–35} These methods are based on the thermodynamic cycle shown in Figure 1. A direct computation of the QM/MM free energy difference $\Delta G_{L_0 \rightarrow L_1}^{QM/MM}$ between ligands L_0 and L_1 (the dashed arrow in Figure 1) is usually unaffordable. Instead, it can be computed via an FEP calculation at the MM level ($\Delta G_{L_0 \rightarrow L_1}^{MM}$; the bottom arrow in Figure 1), followed by two additional FEP calculations that convert the MM Hamiltonian to the QM/MM Hamiltonian for the ligands (the vertical arrows in Figure 1).^{23,27,36} The free energy difference between the MM and QM/MM Hamiltonians can be obtained by single-step exponential averaging

Received: July 6, 2018

Published: October 26, 2018

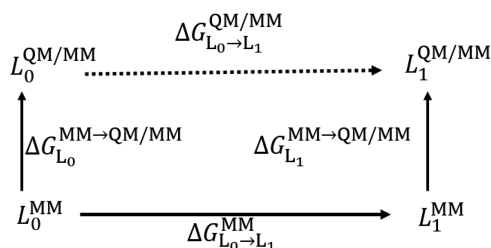


Figure 1. The reference potential approach.

(ssEA)^{27,33,34} and non-Boltzmann Bennett (NBB) calculations.^{32,37} In this way, sampling at the QM/MM level of theory can be avoided.

The prime problem with these approaches is the slow convergence of the free energy difference.^{34,36,38} For instance, the ssEA approach has been shown to give accurate results with low computational effort.^{33,39} However, to obtain converged results for the MM → QM/MM perturbation in a host–guest binding system, ~700 000 QM calculations were needed, and it was necessary to use a cumulant approximation employing interaction energies (and there were indications that the calculations were still not fully converged).²⁷ FEP methods yield accurate results only when the two states are similar enough so that there is sufficient phase-space overlap (i.e., conformations with a large probability in one state also have a high probability to be sampled in the other state). To mitigate this difficulty, FEP calculations are often strengthened by stratification, in which a series of intermediate states are introduced. However, sampling of the intermediate states requires energy and force evaluations at the QM/MM level, making this MM → QM/MM perturbation very demanding.

To solve this problem, Ryde et al. have proposed two alternative approaches: reference potential calculations with QM/MM sampling (RPQS)²⁸ and reference potential calculations with QM/MM sampling and many short simulations (RPQS-MSS).⁴⁰ As the names indicate, both methods require expensive QM/MM sampling. These two methods yielded the same results, and both of them are more efficient than a direct QM/MM FEP calculation (the dashed arrow in Figure 1).²⁸ In the RPQS approach, some intermediate states are introduced to gradually convert the MM Hamiltonian to the QM/MM Hamiltonian. Therefore,

the RPQS approach is much more demanding than ssEA calculations. To enhance the efficiency, multiple short molecular dynamics (MD) simulations are employed in the RPQS-MSS scheme, exploiting the fact that the phase space is already thoroughly sampled at the MM level.⁴⁰ With such an approach, the QM/MM simulation time could be reduced by a factor of 4.

Recently, several studies have shown that nonequilibrium (NE) simulations^{41–49} can be used to calculate the potential of mean force (PMF) along a reaction coordinate^{50–53} and can also be used to estimate binding free energies via alchemical transformations.^{54,55} NE simulations can also be used to estimate the free energy change associated with the MM → QM/MM process, which is a special case of an alchemical transformation.^{56–59} In such simulations, energy and force evaluations with a QM/MM Hamiltonian are still inevitable, but the number of energy and force evaluations can be significantly less than that in a direct equilibrium free energy calculation at the same level.⁴⁵ Moreover, the NE free energy calculations are trivially parallel, which further enhances the efficiency.

NE simulations also employ many short simulations to estimate free energies. Therefore, it is of great interest to compare the efficiencies of the RPQS-MSS and NE methods in the calculation of QM/MM binding free energies. In this work, the relative free energies for the binding of nine cyclic carboxylate ligands to the octa-acid deep-cavity host were calculated at the QM/MM level with the reference potential approach. Nonequilibrium MD simulations were used to estimate the MM → QM/MM free energies. The results are compared to those from our previous studies of the same systems with the QM/MM FEP, RPQS, and RPQS-MSS approaches^{27,28,40} to decide which method is the most efficient.

METHODS

The Reference Potential Approach. In this work, the relative binding free energies were calculated for nine cyclic carboxylate ligands and the octa-acid deep-cavity cavitant, which were taken from the SAMPL4 competition.^{60,61} The octa-acid host and nine guest molecules are shown in Figure 2. Eight ligand transformations were considered in this study, viz.,

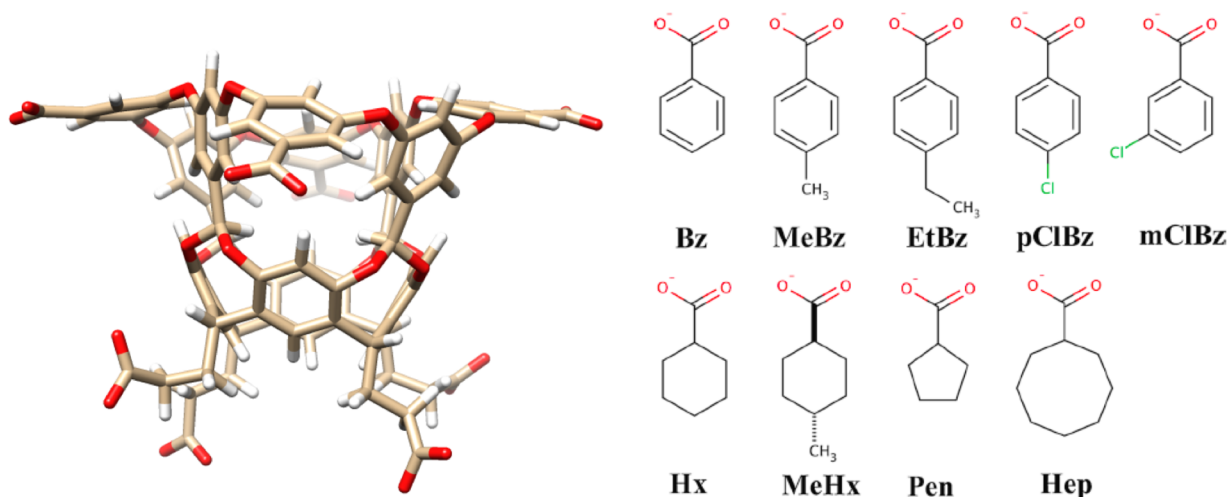


Figure 2. Octa-acid host and the nine guest molecules studied in this work.

MeBz → Bz, EtBz → MeBz, pClBz → Bz, mClBz → Bz, Hx → Bz, MeHx → Hx, Hx → Pen, and Hep → Hx.

QM/MM relative binding free energies were obtained with the reference potential scheme shown in Figure 1. With this thermodynamic cycle, the free energy difference are first determined at the MM level using FEP ($\Delta G_{L_0 \rightarrow L_1, s}^{MM}$, where s refers to either the bound state or the free state) with a series of intermediate states. Then the free energy differences between the MM and QM/MM Hamiltonians ($\Delta G_{L_0, s}^{MM \rightarrow QM/MM}$) are calculated for each ligand (L_0 and L_1). By combining these three results, the free energy difference at the QM/MM level is obtained as

$$\Delta G_{L_0 \rightarrow L_1, s}^{QM/MM} = \Delta G_{L_0 \rightarrow L_1, s}^{MM} - \Delta G_{L_0, s}^{MM \rightarrow QM/MM} + \Delta G_{L_1, s}^{MM \rightarrow QM/MM} \quad (1)$$

Finally, the net binding free energy is obtained by subtracting the free energy difference in the free state from that in the bound state:

$$\Delta G_{L_0 \rightarrow L_1}^{QM/MM} = \Delta G_{L_0 \rightarrow L_1, \text{bound}}^{QM/MM} - \Delta G_{L_0 \rightarrow L_1, \text{free}}^{QM/MM} \quad (2)$$

In this paper, we will primarily discuss the MM → QM/MM free energies, in particular,

$$\Delta G_{L_i}^{MM \rightarrow QM/MM} = \Delta G_{L_i, \text{bound}}^{MM \rightarrow QM/MM} - \Delta G_{L_i, \text{free}}^{MM \rightarrow QM/MM} \quad (3)$$

The MM calculations were taken from our previous studies on the same systems.³⁶ In these simulations, the general Amber force field (GAFF)⁹ was applied for both the host and guest molecules, and the partial charges of the system were obtained by the restrained electrostatic potential (RESP)^{62,63} charge fitting method using the electrostatic potential computed at the HF/6-31G* level of theory. Water molecules were modeled using the TIP3P potential.⁶⁴

Nonequilibrium Simulations. In this work, NE simulations were applied within the reference potential with QM/MM sampling scheme to calculate the free energy difference between the MM and QM/MM Hamiltonians, an approach that will be denoted as RPQS-NE hereafter. In 1997, Jarzynski proved that an equilibrium free energy difference can be expressed as an exponential average of the work performed on the system during an irreversible transformation between two states,^{41,42} viz.,

$$\exp(-\beta \Delta G) = \langle \exp(-\beta W) \rangle \quad (4)$$

where the angle brackets $\langle \dots \rangle$ denote the average over an ensemble of N_{trj} nonequilibrium transformation processes initiated from equilibrium states and $\beta = 1/RT$, where R is the gas constant and T is the absolute temperature ($T = 300$ K in this study). The nonequilibrium work W is accumulated as⁴³

$$\begin{aligned} W(t + \delta t) &= W(t) + \delta W(t \rightarrow t + \delta t) \\ &= W(t) + U(\mathbf{r}(t), \lambda(t + \delta t)) - U(\mathbf{r}(t), \lambda(t)) \end{aligned} \quad (5)$$

where $W(t)$ and $W(t + \delta t)$ are the work done to the system at time t and $t + \delta t$, respectively, δW is the work associated with switching the Hamiltonian from $H(\lambda(t))$ to $H(\lambda(t + \delta t))$ with the coordinates fixed at $\mathbf{r}(t)$, and U is the potential energy of the system, which is a function of both the coordinates and λ . Thus, the work in eq 4 can be obtained from

$$W = \sum_{t=0}^{\tau-\delta t} U(\mathbf{r}(t), \lambda(t + \delta t)) - U(\mathbf{r}(t), \lambda(t)) \quad (6)$$

If not otherwise stated, 100 independent NE simulations were used to calculate the free energy difference between the MM and QM/MM Hamiltonians for each ligand, either bound to the receptor or free in water (i.e., $N_{\text{trj}} = 100$). The QM calculations were performed with the semiempirical PM6-DH+ method,⁶⁵ and only the ligand was included in the QM region, whereas the host molecule was described with GAFF. The TIP3P water model was utilized for the solvent water molecules. The PM6-DH+ method was selected because our previous calculations were performed at this level of theory^{28,40} and it is among the best semiempirical methods available in the Amber software. All of the simulations were carried out with the QM/MM approach⁶⁶ implemented in the sander module of the Amber14 software⁶⁷ using the dual-topology scheme.

The NE simulations were started from the coordinates and velocities saved during the production simulation in a canonical ensemble for the two end states at the MM level. The temperature was kept at 300 K using Langevin dynamics,⁶⁸ and the pressure was set to 1 atm using Berendsen's barostat.⁶⁹ The nonbonded interactions were truncated in real space at 8.0 Å, but the long-range Coulomb interactions were calculated using the particle mesh Ewald method.⁷⁰ No bond length constraints were used, and the time step in the simulations was 1 fs. For each NE simulation, the potential was changed from MM to QM/MM by employing the coupling parameter λ :

$$U(\lambda) = (1 - \lambda)U_{\text{MM}} + \lambda U_{\text{QM/MM}} \quad (7)$$

If not otherwise stated, in each simulation λ was changed from 0 to 1 in 200 stages with an interval of 0.1 ps between two adjacent jumps of λ . Therefore, the total simulation time for each trajectory was 20 ps. In Amber, this is determined by the parameters $\text{dynlmb} = 0.005$ and $\text{ntave} = 100$ (λ increases by 0.005 every 100 MD steps). In what follows, these two parameters are denoted as $\Delta\lambda$ and N_{tsr} , respectively.

Standard Error and Convergence Criteria. The precision of the free energies from the NE simulation was estimated according to⁷¹

$$\sigma^2 = \frac{1}{\beta^2 N_{\text{trj}}} \left(\frac{\langle \exp(-\beta W)^2 \rangle}{\langle \exp(-\beta W) \rangle^2} - 1 \right) \quad (8)$$

It was estimated using numerically stable algorithms to avoid overflow and rounding errors. We confirmed that bootstrapping gave the same standard error within 0.1 kJ/mol (cf. Table S1).

To study the convergence of the MM → QM/MM free energy difference, five criteria were employed. The first criterion was the standard deviation of the energy difference distribution σ , as recommended by Ryde.⁷² The second one was the reweighting entropy S_w :³⁴

$$S_w = -\frac{1}{\ln N_{\text{trj}}} \sum_{j=1}^{N_{\text{trj}}} w_j \ln w_j \quad (9)$$

where the weights w_j are given by

$$w_j = \frac{\exp(-\beta W_j)}{\sum_{i=1}^{N_{\text{trj}}} \exp(-\beta W_i)} \quad (10)$$

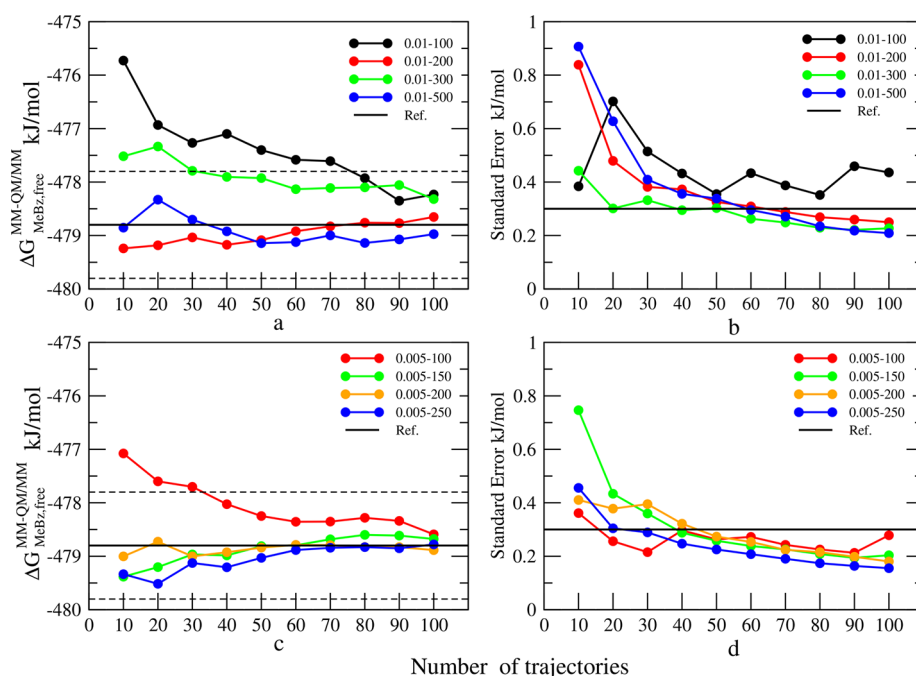


Figure 3. Convergence of (a, c) $\Delta G_{\text{MeBz,free}}^{\text{MM} \rightarrow \text{QM/MM}}$ and (b, d) the standard error with respect to N_{trj} for (a, b) $\Delta\lambda = 0.01$ and (c, d) $\Delta\lambda = 0.005$ and different values of N_{ts} (for each curve, the legend indicates $\Delta\lambda - N_{\text{ts}}$). Curves with the same color have the same total length of the simulation. Black dashed horizontal lines denote deviations of ± 1 kJ/mol from the reference value for $\Delta G_{\text{MeBz,free}}^{\text{MM} \rightarrow \text{QM/MM}}$ (shown by the black solid horizontal lines).

The third one was the maximum value of these weights, w_{max} which provides the largest contribution to the exponential average. The fourth one was Kish's effective sampling size, which is defined as

$$Q = \frac{\left(\sum_{j=1}^{N_{\text{trj}}} w_j\right)^2}{\sum_{j=1}^{N_{\text{trj}}} w_j^2} \quad (11)$$

To make this quantity independent of the actual number of samples, $Q' = Q/N_{\text{trj}}$ was used. The last one was the bias metric Π defined by Wu and Kofke,⁷³ which can be calculated from

$$\Pi = \sqrt{W_L \left[\frac{1}{2\pi(N_{\text{trj}} - 1)^2} \right]} - \sqrt{2\beta(\langle w \rangle - \Delta G)} \quad (12)$$

where $W_L(x)$ is the Lambert W function.

RESULTS AND DISCUSSION

We studied whether nonequilibrium MD simulations with Jarzynski's equality can be employed to calculate accurate QM/MM binding free energies. Toward this end, we studied the binding of nine cyclic carboxylate ligands to the octa-acid deep-cavity host.^{60,61} We used the semiempirical PM6-DH+ method for the ligand,⁶⁵ the GAFF force field for the host,⁹ and TIP3P for water molecules.⁶⁴ We previously studied this system with several other QM/MM-FEP approaches (ssEA, full QM/MM-FEP, RPQS, and RPQS-MSS),^{27,28,40} giving firm reference values and allowing us to discuss the efficiencies of the various methods. The RPQS calculations employed the same thermodynamic cycle as in the present RPQS-NE simulations (Figure 1), but $\Delta G_{\text{L},s}^{\text{MM} \rightarrow \text{QM/MM}}$ was calculated by FEP with four to nine Λ values, employing 1.5 ns QM/MM MD simulations for each Λ value. RPQS-MSS used four Λ

values and five to 43 simulations of 4×20 ps to reach a precision of 0.3 kJ/mol.

Parameters for the NE Simulations. Optimal control parameters for the NE simulations had to be determined first. To this end, the $\text{MM} \rightarrow \text{QM/MM}$ free energy difference of MeBz in the free state was studied. This molecule was chosen because it showed quite slow convergence in our previous study.⁴⁰ As mentioned in **Methods**, the accuracy and time consumption of the NE simulations depend on three parameters: the number of NE simulations to compute the average (N_{trj}), the number of MD steps between each jump in λ (N_{ts}), and the change in λ in each jump ($\Delta\lambda$). Thus, the total number λ values used is $1/\Delta\lambda$, and since we used a time step of 1 fs, the total time (in ps) for each individual simulation in the ensemble is $\frac{N_{\text{ts}}}{1000\Delta\lambda}$. As will be discussed further below, we used N_{trj} to obtain the desired precision of the calculated free energies, but proper values of the other two parameters had to be determined.

Some initial test calculations (cf. the **Supporting Information**) showed that simulations with $\Delta\lambda = 0.01$ and 0.005 gave a proper compromise between accuracy and time consumption. The results of these two choices for $\Delta\lambda$ are presented in Figure 3 with varying N_{ts} (100–500) and N_{trj} (10–100). The reference values (solid lines in the figure) were obtained from our previous study with full RPQS and four Λ values.²⁸ It can be seen that all of the estimated $\Delta G_{\text{MeBz,free}}^{\text{MM} \rightarrow \text{QM/MM}}$ converged to within 1 kJ/mol (shown by the dashed horizontal lines) of the reference value (shown by the solid horizontal lines) when N_{trj} is large enough. This convergence limit was also employed in our previous RPQS-MSS study and represents both a reasonable level of accuracy and an approximate 95% confidence interval for the difference between the RPQS and RPQS-NE results.⁴⁰ However, the standard error (SE) of the shortest simulation, 0.01–100 (i.e., with $\Delta\lambda = 0.01$ and $N_{\text{ts}} = 100$, shown in black) is larger than that of the reference

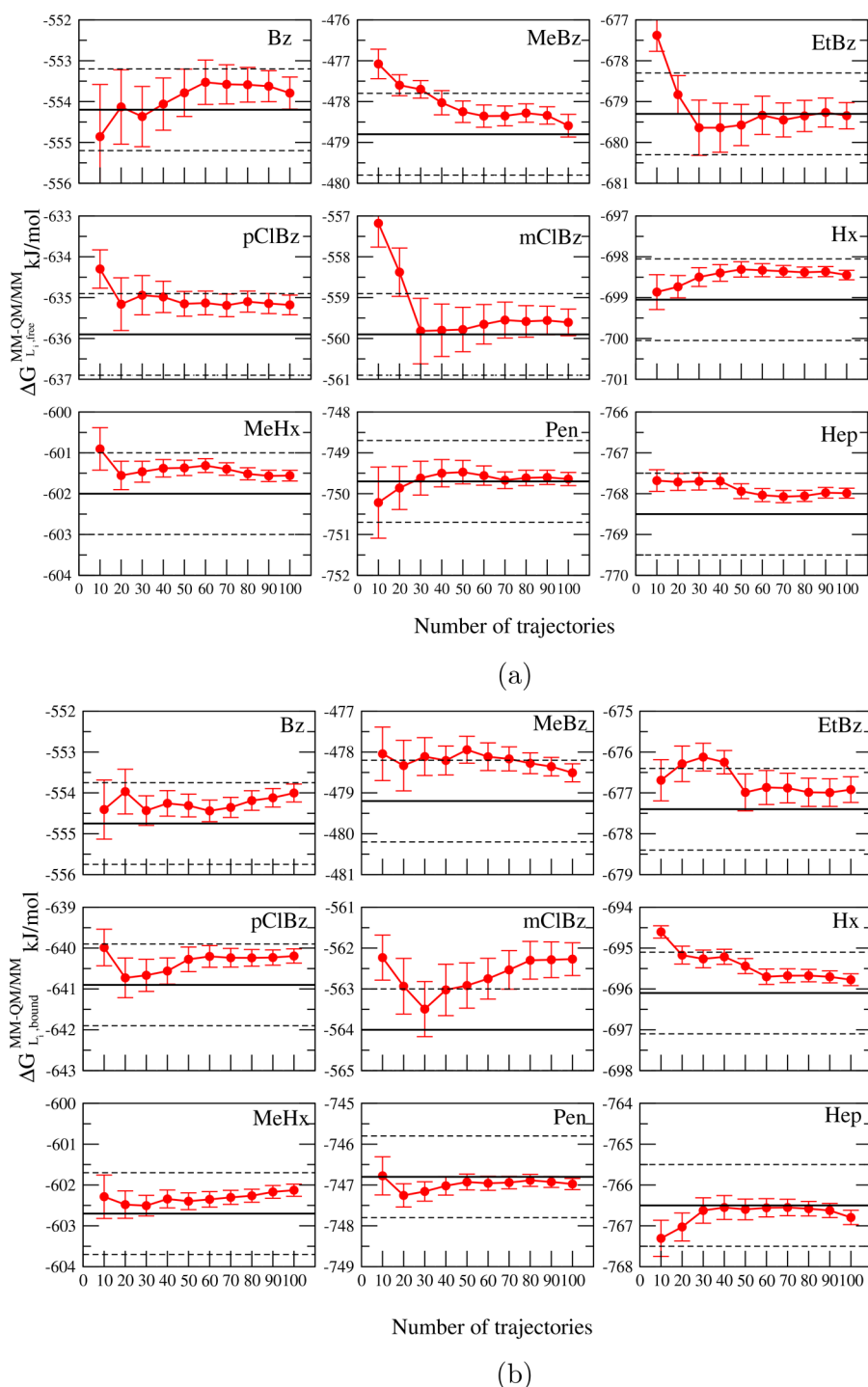


Figure 4. Convergence of (a) $\Delta G_{L, \text{free}}^{\text{MM} \rightarrow \text{QM/MM}}$ and (b) $\Delta G_{L, \text{bound}}^{\text{MM} \rightarrow \text{QM/MM}}$ as functions of N_{trj} . The RPQS results with 1.5 ns MD simulations for four Δ values are shown as black solid horizontal lines, with deviations of ± 1 kJ/mol shown as black dashed horizontal lines.

simulation (0.3 kJ/mol, cf. Figure 3b) and does not show the expected $1/\sqrt{N_{\text{trj}}}$ behavior. The second-shortest simulations are the 0.01–200 and 0.005–100 combinations, shown in red in Figure 3. Both simulations give converged $\Delta G_{L, \text{free}}^{\text{MM} \rightarrow \text{QM/MM}}$, but the SE is lower and the convergence is faster for the 0.005–100 simulation (the 0.01–200 simulation requires 60 trajectories before the SE is below 0.3 kJ/mol). It can also be seen that the results are well-converged compared with longer simulations (green, yellow, and blue curves in Figure 3).

Therefore, the combination of $\Delta\lambda = 0.005$ and $N_{\text{ts}} = 100$ was chosen for the calculations with the other ligands in this study.

Convergence of $\Delta G_{L, \text{free}}^{\text{MM} \rightarrow \text{QM/MM}}$ with Respect to the Number of Trajectories N_{trj} . Next, we studied how $\Delta G_{L, \text{free}}^{\text{MM} \rightarrow \text{QM/MM}}$ and $\Delta G_{L, \text{bound}}^{\text{MM} \rightarrow \text{QM/MM}}$ converge with respect to the number of NE trajectories N_{trj} (using $\Delta\lambda = 0.005$ and $N_{\text{ts}} = 100$) for all the ligands. The results are shown in Figure 4.

As shown in Figure 4a, $\Delta G_{L, \text{free}}^{\text{MM} \rightarrow \text{QM/MM}}$ for all of the free ligands in water converged to the reference values within 1 kJ/mol. For Bz, Hx, Pen, and Hep, the results converged with only

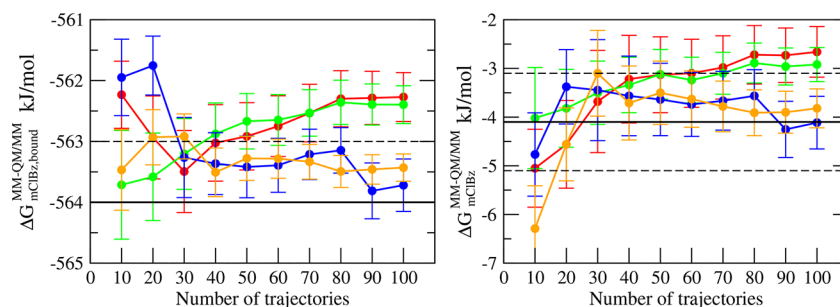


Figure 5. (a) $\Delta G_{\text{mClBz,bound}}^{\text{MM} \rightarrow \text{QM/MM}}$ and (b) $\Delta G_{\text{mClBz}}^{\text{MM} \rightarrow \text{QM/MM}}$ for the mClBz ligand with $N_{\text{ts}} = 100$ (red), 200 (green), 500 (blue), and 1000 (yellow). The reference values are shown as black solid horizontal lines, and the error bars represent the standard errors in the calculations.

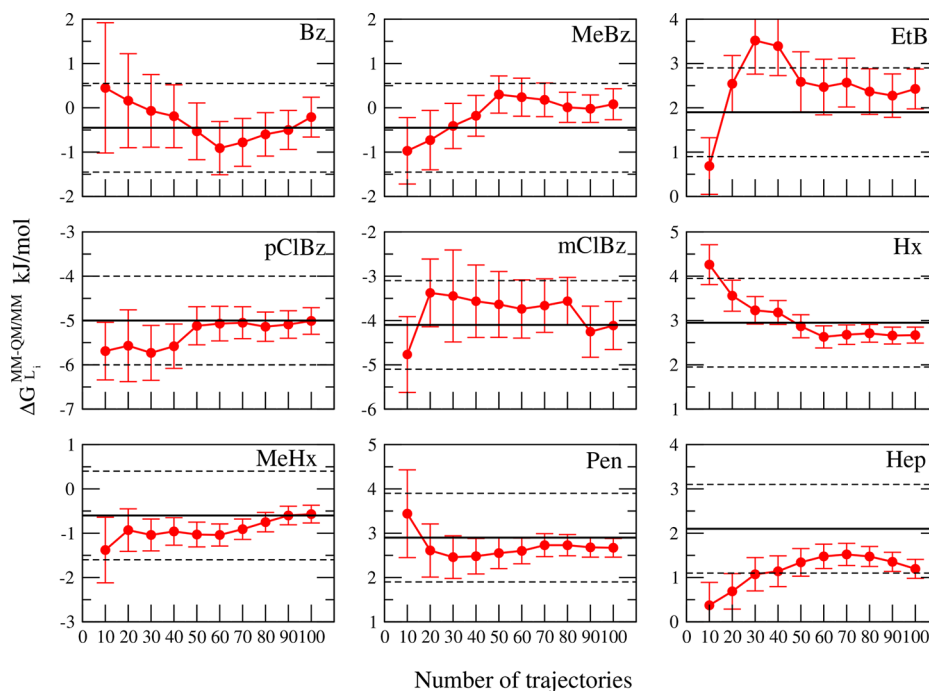


Figure 6. Convergence of $\Delta G_{L_i}^{\text{MM} \rightarrow \text{QM/MM}}$ as a function of N_{trj} . The results were obtained with $\Delta\lambda = 0.005$ and $N_{\text{ts}} = 100$, except for mClBz ($\Delta\lambda = 0.005$ and $N_{\text{ts}} = 500$). RPQS results with four Λ values²⁸ are shown as black solid horizontal lines with deviations of ± 1 kJ/mol marked as black dashed horizontal lines.

10 NE trajectories, and for the other five guest molecules (MeBz, EtBz, pClBz, mClBz, and MeHx), 20–40 trajectories were required to yield converged results. With 100 NE trajectories, $\Delta G_{L_i}^{\text{MM} \rightarrow \text{QM/MM}}$ agreed with the reference values within 0.5 kJ/mol for all of the ligands except pClBz and Hx (0.7 kJ/mol). Moreover, for most of the ligands, the SEs of the calculations are less than or equal to those of the reference simulations (listed in Table S1). The only exceptions were Bz and EtBz, for which the SEs were 0.1 kJ/mol larger than for the reference simulation. For MeBz, pClBz, Hx, MeHx, and Hep, fewer than 30 trajectories were enough to yield the same SE as in the reference calculations.

For the guest molecules bound to the host, $\Delta G_{L_i}^{\text{MM} \rightarrow \text{QM/MM}}$ converged with less than 10 trajectories for five ligands (Bz, pClBz, MeHx, Pen, and Hep), as can be seen in Figure 4b and Table S1. For Hx, the results converged with 20 trajectories, but for EtBz and MeBz 50 and 80 trajectories were required, respectively. However, the deviations from the reference values were somewhat larger for the bound state. Only for Hx, Pen, and Hep were the differences less than 0.5 kJ/mol with 100 trajectories. Nonetheless, the SEs of the bound states were no

larger than in the reference simulations, except for mClBz. For Bz, MeBz, Hx, MeHx, and Pen, 50 trajectories was enough to yield the same SE as in the reference.

However, $\Delta G_{\text{mClBz,bound}}^{\text{MM} \rightarrow \text{QM/MM}}$ for mClBz did not converge to the reference value even after 100 trajectories. This ligand also gave poor convergence in our previous RPQS-MSS study,⁴⁰ and we showed that the problem is caused by the fact that the preferred orientation of the chlorine atom in the host is different in the MM (to the side) and QM simulations (to the bottom of the host). Therefore, we carried out additional simulations, in which N_{ts} was increased to 200, 500, and 1000. The results in Figure 5 show that with $N_{\text{ts}} = 200$ (green line), $\Delta G_{\text{mClBz,bound}}^{\text{MM} \rightarrow \text{QM/MM}}$ was still not converged, giving a final deviation from the reference by 1.6 kJ/mol. However, when N_{ts} was increased to 500 (blue line), convergence within 1 kJ/mol was reached with 30 trajectories. Although the SE was 0.1 kJ/mol larger than the reference value with 100 trajectories, the result seems to be reliable. When N_{ts} was further increased to 1000, $\Delta G_{\text{mClBz,bound}}^{\text{MM} \rightarrow \text{QM}}$ converged to a constant value deviating by only 0.5 kJ/mol from the reference with only 40 trajectories, and

Table 1. Calculated $\Delta G_{L_i}^{MM \rightarrow QM/MM}$ values with Standard Errors (in kJ/mol); For Comparison, the RPQS Results Obtained with Four Λ Values Are Also Shown, and the Last Three Rows Indicate the Numbers of Trajectories Needed To Reach Certain Precisions

N_{trj}	Bz	MeBz	EtBz	pClBz	mClBz	Hx	MeHx	Pen	Hep
10	0.45 $\pm$ 1.47	−0.97 $\pm$ 0.75	0.69 $\pm$ 0.64	−5.69 $\pm$ 0.65	−4.77 $\pm$ 0.86	4.26 $\pm$ 0.45	−1.38 $\pm$ 0.74	3.44 $\pm$ 0.99	0.37 $\pm$ 0.52
20	0.16 $\pm$ 1.06	−0.73 $\pm$ 0.67	2.54 $\pm$ 0.64	−5.57 $\pm$ 0.81	−3.38 $\pm$ 0.76	3.56 $\pm$ 0.35	−0.93 $\pm$ 0.48	2.61 $\pm$ 0.60	0.69 $\pm$ 0.40
30	−0.07 $\pm$ 0.82	−0.41 $\pm$ 0.51	3.52 $\pm$ 0.76	−5.73 $\pm$ 0.62	−3.45 $\pm$ 1.03	3.23 $\pm$ 0.31	−1.04 $\pm$ 0.36	2.46 $\pm$ 0.48	1.07 $\pm$ 0.38
40	−0.19 $\pm$ 0.71	−0.18 $\pm$ 0.46	3.39 $\pm$ 0.66	−5.58 $\pm$ 0.50	−3.56 $\pm$ 0.82	3.18 $\pm$ 0.27	−0.96 $\pm$ 0.31	2.48 $\pm$ 0.40	1.14 $\pm$ 0.35
50	−0.53 $\pm$ 0.64	0.30 $\pm$ 0.42	2.59 $\pm$ 0.68	−5.12 $\pm$ 0.43	−3.64 $\pm$ 0.74	2.87 $\pm$ 0.26	−1.03 $\pm$ 0.28	2.55 $\pm$ 0.35	1.34 $\pm$ 0.31
60	−0.91 $\pm$ 0.60	0.24 $\pm$ 0.43	2.47 $\pm$ 0.63	−5.07 $\pm$ 0.39	−3.74 $\pm$ 0.66	2.63 $\pm$ 0.25	−1.04 $\pm$ 0.25	2.60 $\pm$ 0.29	1.48 $\pm$ 0.28
70	−0.78 $\pm$ 0.54	0.18 $\pm$ 0.38	2.57 $\pm$ 0.55	−5.05 $\pm$ 0.36	−3.66 $\pm$ 0.60	2.68 $\pm$ 0.22	−0.91 $\pm$ 0.23	2.73 $\pm$ 0.26	1.52 $\pm$ 0.25
80	−0.60 $\pm$ 0.49	0.01 $\pm$ 0.34	2.37 $\pm$ 0.52	−5.14 $\pm$ 0.33	−3.56 $\pm$ 0.54	2.71 $\pm$ 0.20	−0.75 $\pm$ 0.22	2.73 $\pm$ 0.24	1.47 $\pm$ 0.23
90	−0.50 $\pm$ 0.44	−0.02 $\pm$ 0.31	2.27 $\pm$ 0.49	−5.09 $\pm$ 0.31	−4.25 $\pm$ 0.58	2.66 $\pm$ 0.19	−0.60 $\pm$ 0.21	2.68 $\pm$ 0.22	1.35 $\pm$ 0.22
100	−0.21 $\pm$ 0.45	−0.08 $\pm$ 0.35	2.43 $\pm$ 0.45	−5.01 $\pm$ 0.30	−4.11 $\pm$ 0.54	2.67 $\pm$ 0.18	−0.57 $\pm$ 0.20	2.67 $\pm$ 0.21	1.19 $\pm$ 0.21
ref ^a	−0.53 $\pm$ 0.38	−0.46 $\pm$ 0.38	1.94 $\pm$ 0.38	−5.03 $\pm$ 0.35	−4.11 $\pm$ 0.43	2.95 $\pm$ 0.35	−0.63 $\pm$ 0.34	2.89 $\pm$ 0.34	2.06 $\pm$ 0.35
N_{MSS}^b	38	25	24	18		9	13	6	13
N_{ref}^c	139	86	143	74	159	26	36	38	36
$N_{0.3}^d$	225	136	225	100	324	36	44	49	49

^aFrom ref 28. ^bNumber of trajectories needed to reach a standard error of 0.3 kJ/mol with RPQS-MSS.⁴⁰ ^cEstimated number of trajectories needed to reach the same standard error as in the reference. ^dEstimated number of trajectories needed to reach a standard error of 0.3 kJ/mol.

the SE decreased to 0.2 kJ/mol after 100 trajectories, which is lower than for the reference calculation.

Finally, the net $\Delta G_{L_i}^{MM \rightarrow QM/MM}$ was computed by combining the results for each ligand in the bound and free states according to eq 3. As shown in Figure 6 and Table 1, the results converged to within 1 kJ/mol of the reference values for all nine guest molecules with 10–50 trajectories. Even with 40 trajectories, the values for seven of the nine ligands (all but EtBz and Hep) converged, and the deviations from the reference values were not greater than 0.5 kJ/mol. For most of the ligands, the SE was lower than in the RPQS study after 20 (Hx and Hep) to 70 (MeBz) trajectories, but for Bz, EtBz, and mClBz, the SE was slightly larger than 0.5 kJ/mol even with 100 trajectories.

We tried to use five convergence criteria (σ , S_w , w_{max} , Q' , and Π) to decide proper values of the parameters $\Delta\lambda$ and N_{ts} . However, as discussed in the Supporting Information, it was hard to find conclusive criteria that worked for all of the systems and gave reliable trends for variations in $\Delta\lambda$ and N_{ts} , although $\sigma < 0.8RT$, $S_w > 0.9$, $Q' > 0.4$, and $\Pi > 1.3$ may give some indication that the parameters are properly selected.

Efficiency. As shown in the previous section, the RPQS-NE method gives the same binding free energies (within statistical uncertainty) as obtained in our previous RPQS and RPQS-MSS studies of the same systems,^{28,40} which is quite satisfactory. Therefore, the next interesting question is which of the methods is the most efficient, i.e., which one requires the smallest computational effort. To answer this question, we compare calculations that gave the same precision (because for all of the methods, the precision can be improved by running more or longer trajectories). For the RPQS and RPQS-NE methods, the SEs are shown in Table 1. In the penultimate row, the number of NE trajectories needed to reach the same SE as in the RPQS trajectories is given (N_{ref}), as estimated from the SE with 100 trajectories and assuming a square-root dependence. It can be seen that 26–159 trajectories are needed, for a total of 736 trajectories for the nine ligands. Since the length of each simulation is 20 ps (100 ps for mClBz), the total simulation time for the nine ligands is 27.4 ns (in fact, the total simulation time is twice as long, because trajectories are needed for both the free and bound states). This should be

compared to the RPQS calculations, which require $9 \times 4 \times 1.5 = 54$ ns (nine ligands, four λ values, and 1.5 ns for each simulation). Thus, the RPQS-NE calculations are twice as effective as the original RPQS calculations. Moreover, excluding the problematic mClBz ligand (which requires the largest number of trajectories, each 5 times longer than for the other ligands), the RPQS-NE calculations are 4 times more efficient.

For the RPQS-MSS calculations, we aimed at an SE of 0.3 kJ/mol (OPT1 in ref 40). The last row in Table 1 shows the number of NE trajectories needed to obtain a SE of 0.3 kJ/mol ($N_{0.3}$, estimated in the same way). It can be seen that 36–225 NE trajectories are needed (excluding mClBz, which was not considered in ref 40), giving a sum of 865 trajectories and a total length of 17.3 ns. For the corresponding RPQS-MSS calculations, only 145 trajectories were needed to reach the same SE (N_{MSS} in Table 1), but each of them had a length of $4 \times 20 = 80$ ps (four Λ values, each with 20 ps of simulation), giving a total of 11.6 ns. This indicates that the RPQS-NE approach is actually 1.5 times more expensive than the RPQS-MSS calculations. Looking at the individual trajectories, RPQS-NE is more efficient for the MeHx and Hep ligands (by a factor of 1.1–1.2), whereas it is less efficient for the other ligands, with the largest difference for EtBz (a factor of 2.3).

CONCLUSIONS

We have investigated whether QM/MM calculations of binding free energies can be accelerated by the reference potential approach using NE simulations and Jarzynski's equality to calculate the $MM \rightarrow QM/MM$ free energy correction. We studied the binding of nine cyclic carboxylate ligands to the octa-acid deep-cavity host, taken from the SAMPL4 challenge,^{60,61} which was previously studied by Ryde and co-workers with many methods, including MM-FEP, direct QM/MM-FEP, ssEA, NBB, RPQS, and RPQS-MSS.^{27,28,36,40} The MM calculations were run with the GAFF force field, whereas the ligand was treated by the semiempirical PM6-DH+ method in the QM/MM calculations.

By comparison to the QM/MM-FEP, RPQS, and RPQS-MSS results (which were obtained with the same QM/MM

approach), we can confirm that the present RPQS-NE approach gives $MM \rightarrow QM/MM$ free energy corrections that agree with the previous results within the statistical uncertainty, i.e., ~ 1 kJ/mol (the mean absolute deviation from the reference RPQS results with four Λ values is 0.4 kJ/mol). Thus, we have shown that RPQS-NE is an accurate approach for the calculation of QM/MM binding free energies.

Both RPQS-MSS and RPQS-NE are rigorous methods in the sense that when provided with enough sampling and λ values they give the correct results. However, if the number of simulations, the simulation length, and the number of λ values or the rate of change in λ are too small, both methods become less accurate. We found that 4.3 ns QM/MM NE trajectories were required per ligand on average to reach a precision of 0.3 kJ/mol. This is 1.5 times more than for RPQS-MSS, so at least for these systems, there does not seem to be any advantage in running NE trajectories rather than many short equilibrium trajectories. However, there may very well be other systems for which the situation is different.

■ ASSOCIATED CONTENT

● Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jctc.8b00685.

Results for $\Delta G_{L,free}^{MM \rightarrow QM/MM}$ and $\Delta G_{L,bound}^{MM \rightarrow QM/MM}$ and convergence criteria obtained for $\Delta G_{MeBz,free}^{MM \rightarrow QM/MM}$ with different values of N_{trj} (PDF)

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Funding

U.R. was supported by grants from the Swedish Research Council (Project 2014-5540) and the Knut and Alice Wallenberg Foundation (KAW 2013.0022). M.W. was supported by a scholarship from the China Scholarship Council. Y.M. was supported by the National Natural Science Foundation of China (Grant 21773066) and the Fundamental Research Funds for the Central Universities. The computations were performed on computer resources provided by the Swedish National Infrastructure for Computing (SNIC) at Lunarc at Lund University.

Notes

The authors declare no competing financial interest.

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